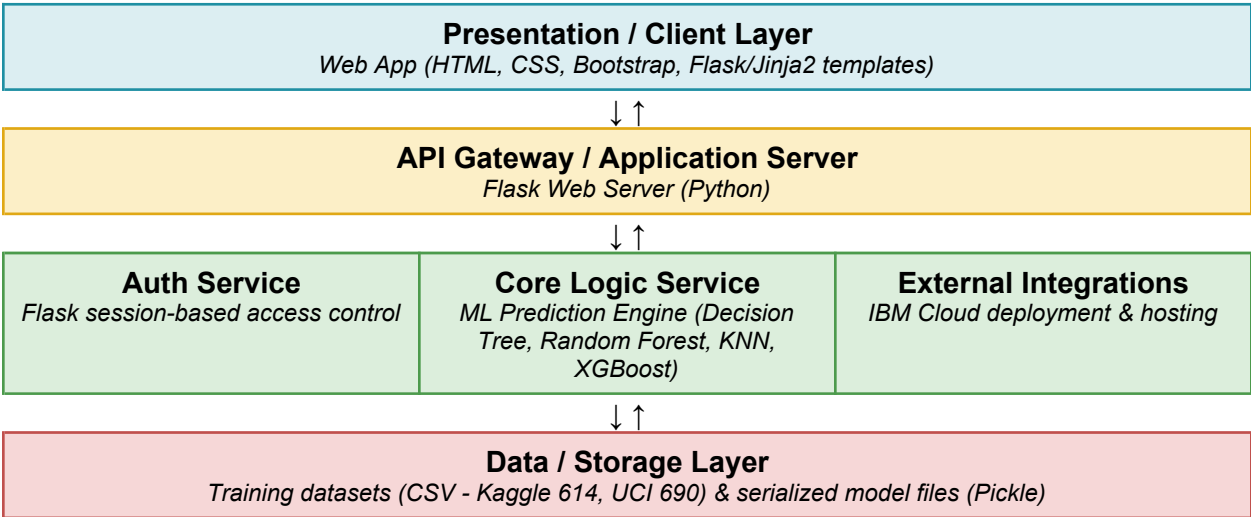


Solution Architecture

Date	30 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level architecture of Smart Lender, mapping the Presentation, Application, and Data layers along with the technologies used at each layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the web interface where applicants enter their details (income, credit history, loan amount, etc.) and view the loan approval prediction result.	HTML, CSS, Bootstrap, Flask (Jinja2 templates)
API Gateway	Receives form submissions from the client, routes requests to the core prediction logic, and returns the prediction response to the user.	Flask (Python)
Microservices / Core Logic	Preprocesses applicant input and runs it through the trained classification model to predict creditworthiness and loan approval likelihood.	Python, scikit-learn, XGBoost, pandas, numpy
Database	Stores the training datasets used to build the models and the serialized best-performing model (XGBoost) used for real-time predictions.	CSV files (Kaggle, UCI datasets), Pickle model file